इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY

सत्यमेव जयते

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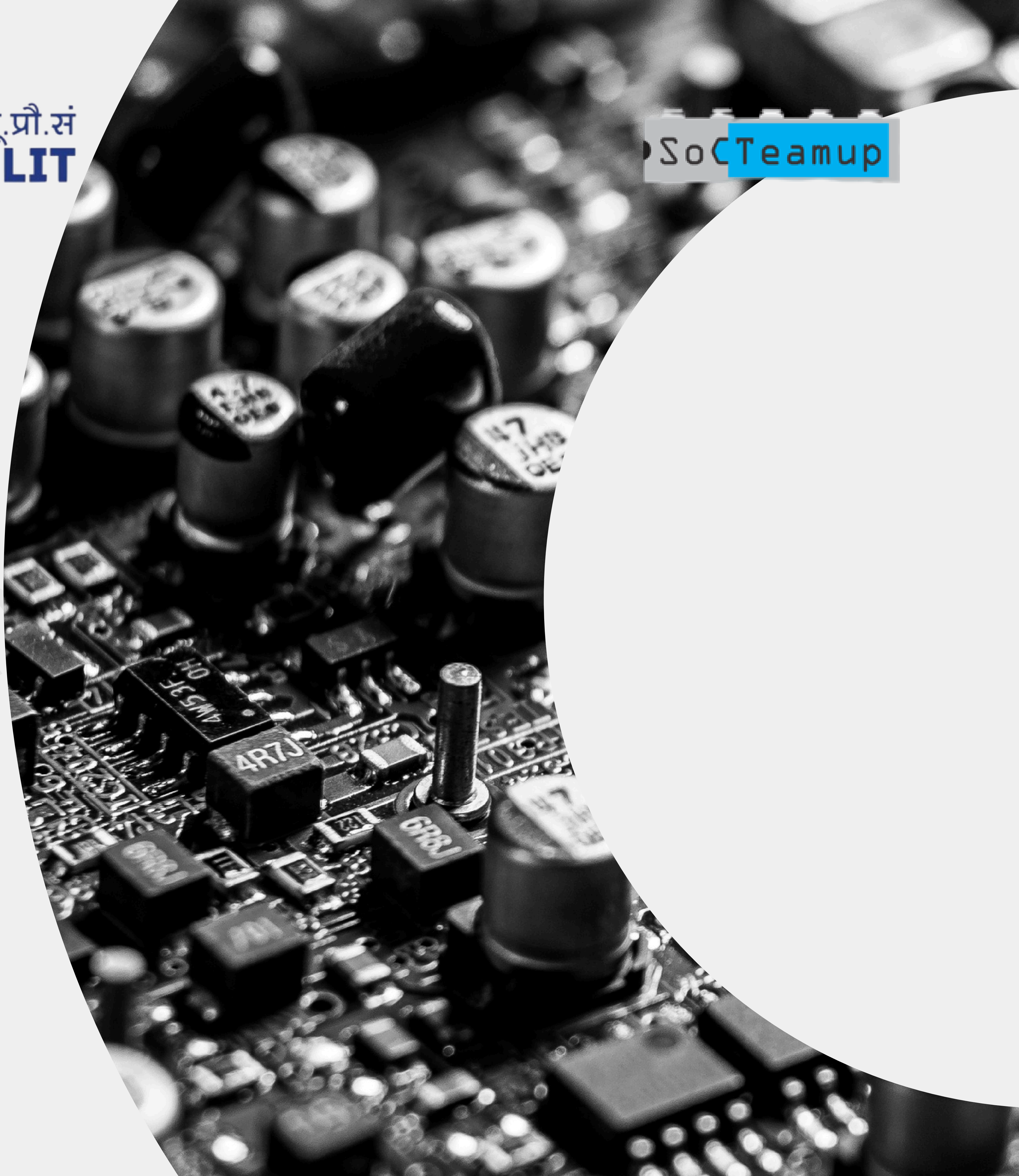
NIELIT

SoCTeamup

Layout placement DEF Overview



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Outline

1	What is a DEF file?
2	Importance of DEF in physical design flow
3	LEF vs. DEF: abstract vs. concrete
4	Key sections in a DEF file
5	Macros, Die Area, Tracks, Rows
6	Component placement and pins
7	Nets and routing overview
8	DEF file snippet example
9	DEF in open-source tools
10	Summary and Best Practices

What is a DEF File?

DEF (Design Exchange Format) is an industry-standard ASCII format used to describe the implementation state of a physical design.

- It captures:
  - Exact instance placement coordinates
  - Floorplan boundary (die/core)
  - Standard cell rows and site definitions
  - I/O pin placements
  - Net connectivity and routing
  - Track grids and via usage
- DEF represents the design-specific layout database, not the technology library.

Importance of DEF in physical design flow

DEF acts as the central data exchange format between:

- Floorplanning
- Placement
- CTS
- Routing
- ECO flows
- Signoff tools

Importance of DEF in physical design flow

It enables:

- Physical consistency across tools
- Incremental implementation updates
- Layout visualization and debug
- Handoff to downstream tools (DRC/LVS, GDS generation)

DEF vs. LEF – What's the Difference?

LEF (Library Exchange Format)	DEF (Design Exchange Format)
Describes the standard cells (abstract physical info)	Describes actual chip layout (concrete placement)
LEF as the **blueprint**	DEF as the **furniture arrangement** on the floorplan.

Key sections of a DEF File

KEY SECTIONS :

VERSION	Format version
UNITS	Distance units
DIEAREA	Bounding box of the chip
ROW	Rows for standard cell placement
COMPONENTS	All cells used and their placement
PINS	I/O locations
NETS	Connectivity
TRACKS and VIAS	Grid info for routing

Example – DIEAREA and ROW

- DIEAREA defines the chip size.
- ROW defines standard cell rows (site, orientation, repeat count).
- DO/BY/STEP defines replication pattern (like grid layout)

```
DIEAREA ( 0 0 ) ( 1200 800 ) ;

ROW ROW_1 coreSite 0 0 N DO 40 BY 1 STEP 30 0 ;

- DIEAREA defines the chip size
  ROW defines standard cell rows (site, orientation, repeat count)
- DO/BY/STEP define replication pattern (like grid layout)
```

Example – COMPONENT Placement

```
COMPONENTS 2 ;
- INVX1 U1 + PLACED ( 100 120 ) N ;
- DFFX1 U2 + PLACED ( 160 120 ) N ;
END COMPONENTS

Each component line tells:
- Cell type and instance name
- Placement coordinates
- Orientation (N = north/facing up)
```

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Components placements and pins

```
PINS 1 ;
- clk + NET clk + DIRECTION INPUT + PLACED ( 0 400 ) N ;
END PINS

NETS 1 ;
- clk ( U1 A ) ( U2 CLK ) ;
END NETS

Defines:
- Location and direction of I/O pins
- Connections between instances and pins
```

Nets and routing overview

```
Tracks define horizontal and vertical metal lines:
TRACKS X 0 DO 100 STEP 4 LAYER M1 ;
TRACKS Y 0 DO 50 STEP 4 LAYER M2 ;

VIA definitions and route guides may also appear.

Helps router know:
- Where wires can legally go
- Via and metal layer transitions
```

DEF file snippet example

```
Tracks define horizontal and vertical metal lines:
TRACKS X 0 DO 100 STEP 4 LAYER M1 ;
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DEF in Open-Source Tools

- Open-source tools like:
  - OpenROAD
  - TritonRoute
  - KLayout
- can read/write DEF files.

DEF in Open-Source Tools

- DEF files are also used in:
  - Floorplan visualization
  - DRC/LVS checks
  - Exporting to GDS

- SkyWater130 PDK provides DEF support.

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Summary and Best Practices

SUMMARY	BEST PRACTICES
DEF files store placement, routing, pin and net info	Keep LEF and DEF matched
Complement LEF to complete physical layout	Check orientation and placement legality
Essential for backend flow (placement, routing, GDS)	Use DEF viewers (KLayout/OpenROAD GUI) to debug layout

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Thank you !